Quality Information Document

Ocean Monitoring Indicator (OMI) of the Copernicus Marine Environment Monitoring Service

Black Sea Ocean Heat Content Anomaly (0-300m)

Issue: 3.0

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CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	16/06/2022	All	Creation of the document	Leonardo Lima	Leonardo Lima
2.0	16/06/2023	All	Updating the time series according to the availability of datasets	Leonardo Lima	Leonardo Lima
3.0	28/05/2025	All	Update the time series based on the results from BLK-REA version BSe4r1 and according to the availability of datasets	Leonardo Lima	E. Peneva, R. Lecci

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I INTRODUCTION

I.1 Product name

OMI_CLIMATE_OHC_BLKSEA_area_averaged_anomalies

I.2 Indicator evaluation method

Ocean heat content is defined here as the deviation from a reference period (1993 -2014) and is closely proportional to the average temperature change from z_1 = 0 m to z_2 = 300 m depth:

$$OHC = \int_{z_1}^{z_2} \rho_0 c_p (T_m - T_{clim}) dz \qquad [1]$$

with a reference density of = 1020 kg m^{-3} and a specific heat capacity of

c_p = 3980J kg^{-1} $°C^{-1}$ (e.g. Lima et al., 2020; von Schuckmann et al., 2009); T_m corresponds to the monthly average temperature and T_{clim} is the climatological temperature of the corresponding month that varies according to each individual product.

Time series of monthly mean values area averaged ocean heat content is provided for the Black Sea (40.86°N, 46.8°N; 27.32°E, 41.96°E) and is evaluated in areas where the topography is deeper than 300m. The Azov and Marmara Seas are not considered.

I.3 CMEMS products used as input for the computations

Product name (and Type)	Documentation
BLKSEA_MULTIYEAR_PHY_006_004	QUID: https://documentation.marine.copernicus.eu/QUID/CMEMS-BLK-QUID-007-004.pdf PUM: https://documentation.marine.copernicus.eu/PUM/CMEMS-BLK-PUM-007-004.pdf

GLOBAL_MULTIYEAR_PHY_001_030	QUID : https://documentation.marine.copernicus.eu/QUID/CMEMS-GLO-QUID-001-030.pdf PUM : https://documentation.marine.copernicus.eu/PUM/CMEMS-GLO-PUM-001-030.pdf
INSITU_GLO_PHY_TS_OA_MY_013_052	QUID : https://documentation.marine.copernicus.eu/QUID/CMEMS-INS-QUID-013-052.pdf PUM : https://documentation.marine.copernicus.eu/PUM/CMEMS-INS-PUM-013-002-052.pdf
MULTIOBS_GLO_PHY_TSUV_3D_MYNRT_015_012	QUID : https://documentation.marine.copernicus.eu/QUID/CMEMS-MOB-QUID-015-012.pdf PUM : https://documentation.marine.copernicus.eu/PUM/CMEMS-MOB-PUM-015-012.pdf

II SCIENTIFIC VALIDATION

II.1 Black Sea Ocean Heat Content Anomaly (0-300m)

More details and full scientific evaluation can be found in Section 2.1 of the Copernicus Marine Ocean State Report issue 2 (OSR2) (von Schuckmann et al., 2018) and Section 2.6 of the Copernicus Marine Ocean State Report issue 4 (Lima et al., 2020). The latter considered the OHC estimation from an ensemble of global reanalysis (e.g. SODA, GLORYS), the regional Black Sea reanalysis and objective analyses (e.g. EN4, CORA), and represents the basis for the present OMI.

The quality evaluation of BLKSEA_OMI_OHC_area_averaged_anomalies follows this "multi-product" approach as introduced in the second issue of the Ocean State Report (von Schuckmann et al., 2018) following the MyOcean's experience (Masina et al., 2017). However, it considers only products that are present in the CMEMS catalog: three global products and a regional (Black Sea) product have been used to build an ensemble mean and its associated ensemble spread (Figure 1, dark red line and shaded area).

Results of Fig. 1 indicate how each member is distributed around the ensemble mean and show a continuous increase of the heat content in the upper 300m over the period 2005-2022 at a rate of 0.77 ± 0.07 W m^{-2}.

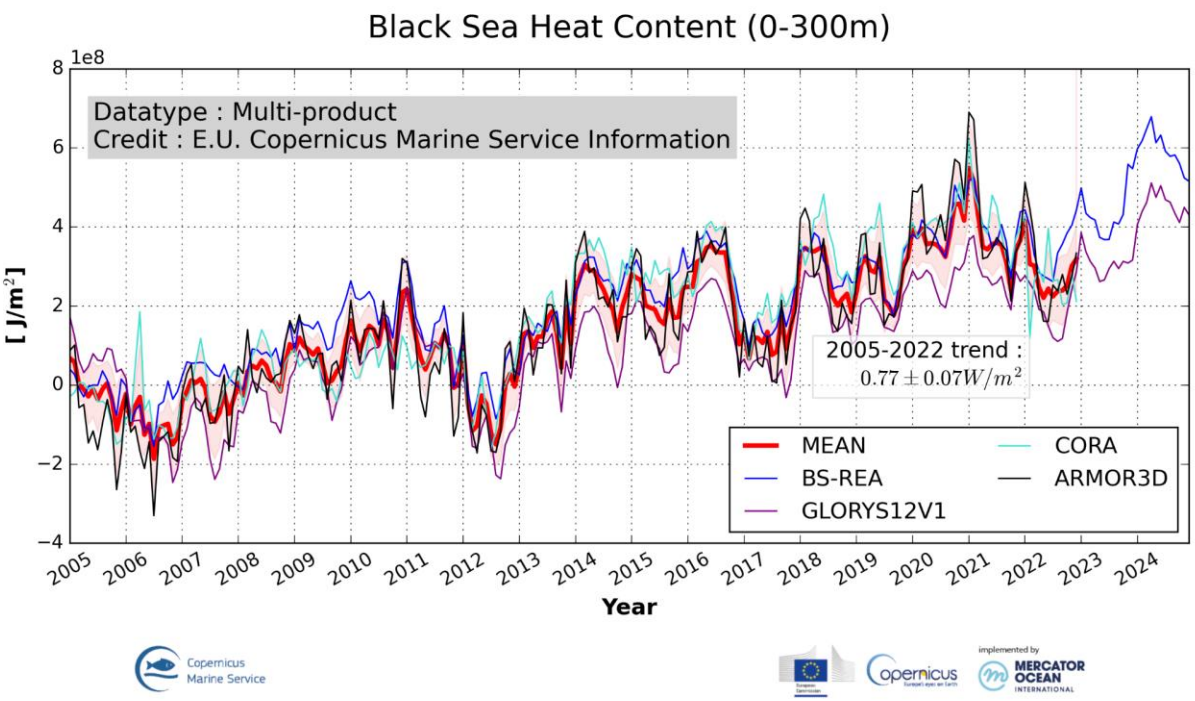

Figure 1:

Time series of area averaged ocean heat content in the Black Sea, and integrated over the 0-300m depth layer. Ocean heat content is defined here as the deviation from a reference period (1993-2014) (Lima et al., 2020). Time series are based on different data products, i.e. Black Sea Reanalysis (dark blue), global ocean reanalysis GLORYS12V1 (purple); global observational based products CORA (light blue), ARMOR3D (black), and the ensemble mean (dark red) and the ensemble spread (red shaded). Details on the products are given in the corresponding PUM for this OMI.

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III REFERENCES

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